

Social Network Analysis: SNAP Facebook Dataset

1. Dataset Overview

- **Dataset Name:** Stanford Network Analysis Project (SNAP) Facebook Social Circles.
- **File Used:** facebook_combined.txt.gz (converted to facebook_edges.csv).
- **Original Network Size:** The initial, unfiltered network consists of **4039 Nodes** and **88,374 Edges**.

2. Key Network Metrics (Centrality)

- **Top 5 Nodes by Degree:**
 - Nodes: 25,30,27,28,0
 - **Interpretation:** These five nodes represent the most "popular" or highly connected individuals in this specific dataset. They have the highest number of direct friendships.
- **Top 5 Nodes by Betweenness Centrality:**
 - Node 25 (Score: 0.0251)
 - Node 30 (Score: 0.0241)
 - Node 27 (Score: 0.0213)
 - Node 28 (Score: 0.0201)
 - Node 0 (Score: 0.0195)
 - **Interpretation:** These nodes act as critical "bridges" or gatekeepers within the network. While they may not have the highest absolute number of friends, they are the most important individuals for keeping different social groups connected.

3. Visual Analysis and Community Structure

- **Layout Algorithm:** The ForceAtlas 2 algorithm was applied to spatialize the network, visually pushing unconnected nodes apart and pulling connected communities together.
- **Community Detection:** By running the Modularity statistic, the network was successfully partitioned into distinct social communities.
- **Interpretation:** The visual clustering shows that Facebook friendships are not randomly distributed. Instead, people group into tight-knit, distinct social circles.

4. Network Filtering (Degree Range (22-163))

- When a topology filter was applied to remove any node with a Degree of range 22-163, the network size reduced from 4,039 nodes to **3,949 nodes** (97.77% visible) and from 88,374 edges to **66,709 edges** (75.48% visible).
- **Interpretation:** This filter effectively removed the "noise" or fringe users who are barely active on the platform. The remaining graph represents the highly active, dense core of the social network.

5. Conclusion: What Was Learned

- **Technical Skills:** Successfully navigated the process of transforming raw text data into a format suitable for Gephi, applying physics-based layout algorithms, and using background statistical calculations (Modularity) to drive visual properties.
- **Analytical Insights:** Learned that identifying the "most important" person in a network depends entirely on the metric used. Degree centrality highlights popularity, but Betweenness centrality highlights structural importance. Visualizing these metrics alongside community detection makes it easy to understand complex social dynamics at a glance.



